

Lecture 4 Part 1:

A Crash Course In

Experimental Design

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Remind of what we did last week

- We spoke about hypothesis and ethics
 - Why we need it, and examples of poor ethical behaviour
 - Importance of ethics within computer science
- We spoke about arguments (philosophically)
 - Inductive vs deductive inference
 - Informal and formal fallacies
 - False dilemmas
 - Straw man arguments
- And we also spoke about possibility vs probability

Contents

- Overview of the video content A crash course in experimental design
- Discussing experimental designs — by example

Video: Experimental Conditions & Variables

Variables and Factors in Experimental Design

Know what the following are:

- Experimental Objects
- Tasks / Processes
- Levels and Conditions
- Independent / Dependent Variables
- Confounding Factors
- Control Variables
- Random Variables

You need to know what these are, and be able to identify them in an experimental design

Variables and Factors in Experimental Design

Need to know these because:

- This is the language of experimental design
- A methodological way of reporting your experiment
- Enables precise critiques of experimental designs, e.g. “why isn’t X a confounding variable” is better than “seems like X might be important for some reason”

You need to learn the language of any topic you work in - this is some standard jargon across all of research regardless of topic

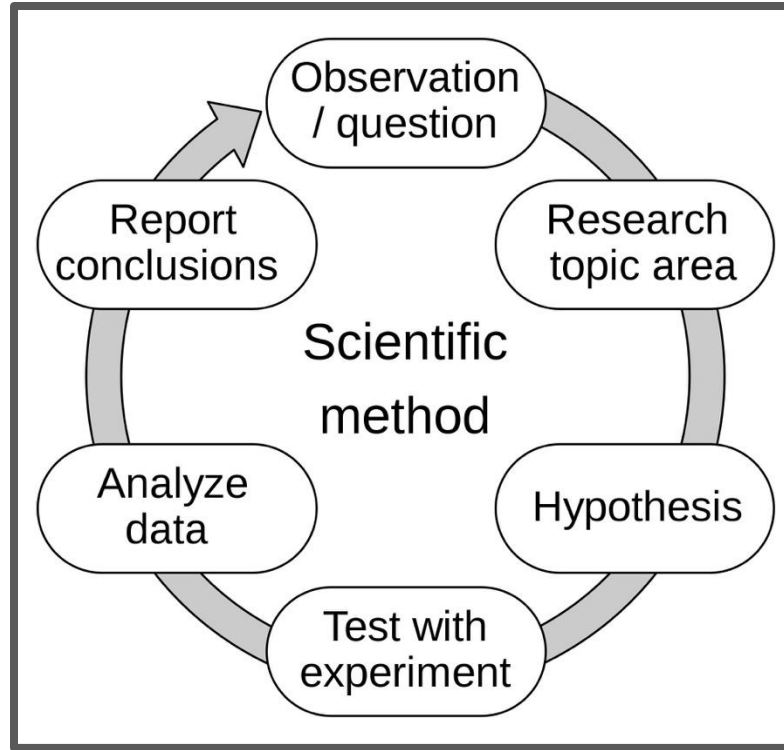
Video: Experiments In Computer Science

VIDEO — Experiments in CS

- Give you an overview of the many types of CS research (going on here at Glasgow)
- Introduce **between** and **within subjects** experimental design

The Scientific Method & Experimental Design

Previously we introduced the scientific method



And the creation of a “hypothesis”

Hypothesis: a **specific** and **testable** explanation for a phenomena.

- *Specific*: explains **what you expect to happen** in a certain circumstance;
- *Testable*: is **falsifiable** (can be disproven by empirical data);
- *Parsimonious*: is as **simple** as feasible (remember **Occam's Razor**);
- *Predictive*: makes a **prediction** that can be tested.

Today we will look at how to
experimentally investigate the
hypothesis

Experimental Design

Experiments

A key part of the scientific method is testing your hypotheses

- Experiments are used to gather empirical data/measurements

We need to use carefully controlled experiments

Experiment designs vary — all serve the same purpose: gathering data against which we can test our hypotheses

The experimental factors checklist

- Research Question / Hypothesis
- Independent / Dependent Variables
- Control Variables Random Variables
- Confounding Factors Participants
- Within/Between Subject Designs

Need to think about all of these aspects – we will cover these today

Independent Variables

Independent variable: what you deliberately change in the experiment

- A variable that the experimenter controls
- Have different “levels”
 - E.g. configuration 1, configuration 2, configuration 3, etc.
- If multiple independent variables — a “condition” is a combination of independent variable levels

Independent Variables

What do you mean by levels of independent variable?

- Each test condition

Example: If “Input Device” is the independent variable examples of levels might be “mouse” and “trackpad” and “eye-gaze”, etc.

Dependent Variables

Dependent variable: the data you are collecting

- A variable whose value(s) depend on the independent variable

Any observable, measurable behaviour can be a dependent variable

- Task execution time, typing speed, accuracy, error rate, questionnaire responses, comfort, emotional response, etc.

Independent and Dependent Variables

- When designing an experiment, an experimenter needs to:
 - Identify 1 or more independent variables
 - These are what the experimenter is going to manipulate
 - Identify 1 or more dependent variables
 - These are the data that are going to be measured / collected

What do we want an experiment to show?

A **change** in the values of the **independent variable** will result in a **change** in the values of the **dependent variable**

We want to be sure that:

- **Any change** in the value of the **dependent variable** is a result of the change in the value of the **independent variable**
(rather than a result of change in the value of other factors)

Control Variables

Control Variables: factors that do not change to ensure a “fair” comparison

- These are not under investigation (i.e. are not IVs) but they might influence participant behaviour (DVs)
- Experimenters control these variables to prevent their influence by setting up their study in a controlled environment

Random Variables

Random Variables: factors you knowingly allow to randomly change/vary

- Instead of attempting to control for everything, it is often better to allow some variables to vary randomly to generalise the results (and because controlling for everything is probably impossible)
 - E.g. Height, finger size, hand size, etc.

Each study requires judgement about the trade off between control and allowing random variation

Confounding Factors

Confounding Factors: factors that change and which impact the results (usually only identified after you have collected the data)

- Factors which could have influenced the relationship between the independent and dependent variable

Confounding Factors

In some cases, a circumstance or condition will change systematically with the independent variable

- E.g. practice, different types of measurement for levels of the IV, prior experience with an interface (keyboard example)

QWERTY KEYBOARD															
~	1	@	#	\$	%	^	&	*	(	)	-	+	=	Delete	
Tab	Q	W	E	R	T	Y	U	I	O	P	{	}		\	
Caps	A	S	D	F	G	H	J	K	L	:	"	'	Enter		
Shift	Z	X	C	V	B	N	M	<	>	?	/	Shift			
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DVORAK Keyboard Layout															
	1	@	#	\$	%	^	&	*	(	)	-	+	=	Delete	
Tab	?	<	>	P	Y	F	G	C	R	L	{	}		\	
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Confounding Factors

Such variables are confounding because they prevent the possibility of a cause-and-effect relationship being inferred from the results

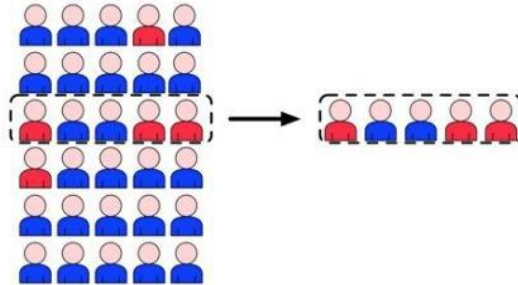
- So, they need to be controlled for
- This is a central skill in experimental design

Participants

In order to correctly assume that research results apply to people other than those recruited you must:

Recruit people from the population you want to investigate

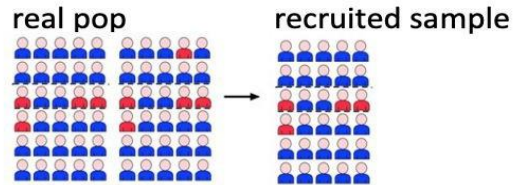
Recruit a sufficient number of participants



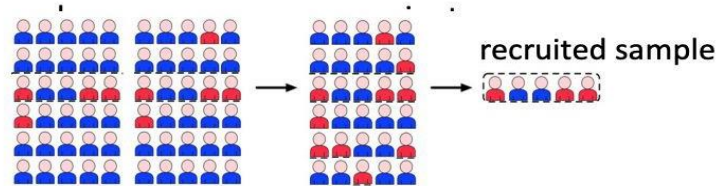
Participants

Recruitment methods

Ideal = participants drawn at random from a population



In practice = convenience sampling



How might these people differ from the population you are trying to generalise your findings to?

Participants

How many people should you recruit?

More is better

However there is a balance between representing the population and practical, and sometimes ethical considerations

Practical: Not much time to recruit (e.g. in student project), population difficult to access, more testing delays product going to market

Ethical: Study puts participants under burden; continuing study beyond necessary delays useful intervention or technique that can improve access.

A note about human characteristics

Since we are talking about user experiments

For user studies (experiments involving human participants) differences in human characteristics are factors in experimental design:

- Age, gender, height, weight, left/right-handed, personality traits, political viewpoints, ethnicity, cultural background, first language, etc.

Since we are talking about user experiments

Variety of ways to handle this:

- Analyse the data separately for differences in human characteristics
- For instance, if you think “age” is a significant factor in your experiment you might run an analysis to compare participants above/below some cutoff age
 - For example, children, older adults etc.

Since we are talking about user experiments

Variety of ways to handle this:

- Deliberately specify such characteristics as random variables
- Typically, with suitable acknowledgement of possible limitations of this on your work
 - For instance, cultural norms differ in the East vs West — but often as researchers we just comment that our research does not look at this.
 - In HCI papers for example, you might see this in a limitation section at the end of a paper.

Since we are talking about user experiments

You might have to change your questions based on participants:

- For example, rephrasing questions to be age appropriate if conducting an experiment with children, such as using the smileometer



Within / Between Subject Designs / Counterbalancing

We need a to, structurally, control how we conduct an experiment to ensure a fair comparison

- For example, to mitigate things like ordering effects, learning, etc.
- The two most common in HCI are:
 - Within-subjects designs and between-subjects designs

Within-Subjects Design

Within-subjects design:

- All subjects are exposed to **all conditions**;
 - i.e., they experience **every level** of the **independent variable(s)**;
 - Subjects are the things you are studying: people, images, search terms, etc.
- Also known as: **repeated-measures** design;
 - Because you repeat the measurements for every condition!

Between-Subjects Design

Between-subjects design:

- Subjects are placed into **groups** and exposed to just **one condition**;
 - i.e., they experience one level of the **independent variable(s)**;
- Also known as **between-groups** or **independent-measures** design;
 - Because you make independent measurements for each condition.

Example:

Take the example comparing the two keyboard types:

QWERTY KEYBOARD

~	1	2	3	4	5	6	7	8	9	0	-	=	Delete
Tab	Q	W	E	R	T	Y	U	I	O	P	[	]	\
Caps	A	S	D	F	G	H	J	K	L	;	'	Enter	
Shift	Z	X	C	V	B	N	M	<	>	?/	Shift		
Ctrl		Alt									Alt		Ctrl

DVORAK Keyboard Layout

	1	2	3	4	5	6	7	8	9	0	-	=	Delete
Tab	?	<	>	P	Y	F	G	C	R	L	[	]	\
	A	O	E	U	I	D	H	T	N	S	'	Enter	
Shift	:	Q	J	K	X	B	M	W	V	Z	Shift		
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ComputerHope.com

You have 20 willing participants for a study, a typing task and way of calculating typing speed and accuracy. You could approach the comparison in two ways:

Within-subjects (repeated measures)

All 20 participants use both keyboards and you compare the average difference in performance between the two.

Between-subjects

You split the participants into two groups of 10 and compare performance of QWERTY group to Dvorak group

Within vs Between Subjects Design

Within-subjects experiments are **less common in other fields**;

- Example: in Medicine, the participants cannot take the treatment drug and the control drug at the same time or in sequence;
- **Between-subjects** designs are typically necessary in other fields because the subjects **can only be exposed to one** of the conditions.

This is usually not an issue in Computing Science;

- Our 'subjects' are often digital, so we can reuse their **initial state**;
 - E.g., we can take the same set of objects and give it to **multiple sort algorithms**;
- But there are exceptions: manipulating real-time data, studies with human participants, 'one-off' events, etc.

Within vs Between Subjects Design

Within-subjects experiments are typically **fairer**;

- Because subjects are exposed to all conditions;
- Contrast with **between-subjects**, where you compare measurements from **different subjects** instead;

Within-subjects experiments are typically **cheaper**;

- If recruiting human participants, every person experiences **all n conditions**, rather than having to hire **n times as many people**;

Within-Subjects Design

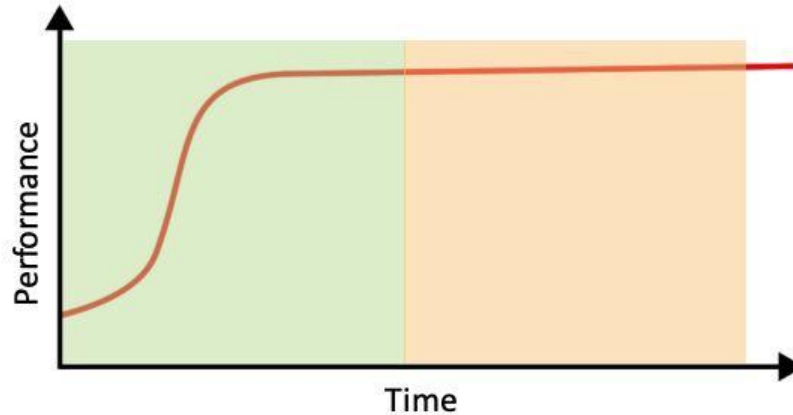
Order effects are a potential limitation to **within-subject** designs:

- When the **order of conditions** has a potential **impact on the DVs**;
- Most common in experiments with human participants;

Learning effects are a particular type of order effect:

- When participants gain knowledge and experience that benefits the conditions towards the end of the experiment;
- Example: participants achieve higher performance later in an experiment because they have **learned how the user interface works**;
- Example: participants perform a programming exercise better towards the end of the experiment because of **experience with earlier tasks**.

Within-Subjects Order Effects



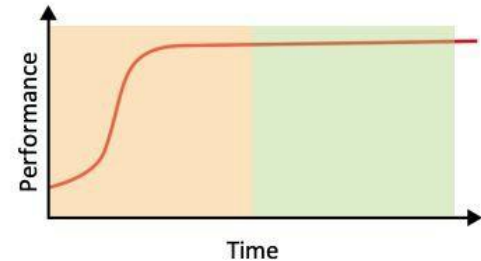
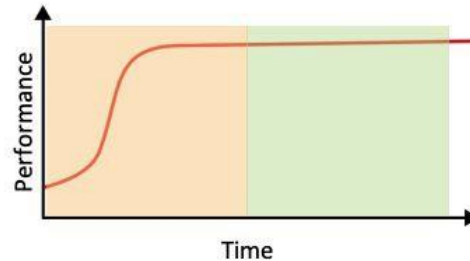
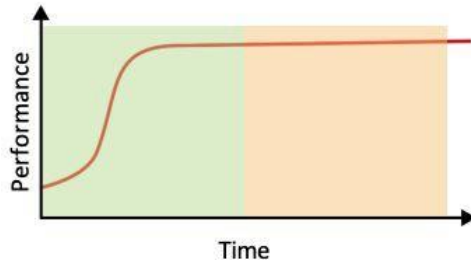
Learning occurs during the **first condition**, so performance will naturally be better during the **second condition**;

- The variance in our dependent variable is more likely to be caused by learning effects than differences in the independent variable(s).

Within-Subjects Order Effects — Randomisation

Randomising condition order helps mitigate order effects:

- Conditions are presented in a random order for each participant;
- By randomising them, order effects get **distributed across conditions**;
 - In an attempt to 'cancel' the effects out.
- But is this optimal?
 - Depends too much on the randomiser...



Counterbalancing — Latin Squares

Counterbalanced designs are an even better alternative:

- Instead of hoping the random order cancels out learning effects, counterbalanced designs are **systematically balanced**;

It would be impractical to use every unique condition ordering;

- For n conditions are $n!$ orders;
 - E.g., four conditions means 24 condition orders.

Instead, use a **Latin Square**: an $n \times n$ array where each symbol occurs only once in each row and once in each column;

- A practical compromise: yields n condition orders such that each condition will appear just once in each position in the sequence.

Counterbalancing — Latin Squares

Order 1

A

B

C

D

Order 2

B

C

D

A

Order 3

C

D

A

B

Order 4

D

A

B

C

Latin Square: an $n \times n$ array where each symbol occurs only once in each row and once in each column;

Finally, it's important to know how to define an experiment

- Given a problem statement you should be able to propose an experimental design to answer it:

Propose an experiment to determine which of two new keyboards is "better"

The experimental checklist

- Research Question
- Criteria
- Independent / Dependent Variables
- Conditions
- Tasks (or Processes)
- Experimental Objects
- Control Variables
- Random Variables
- Possible Confounding Factors

First — what is better?

Better could mean any number of things:

- Latency of input
- Speed which a user can use it to type
- Accuracy which users can use it whilst typing
- User experience using the device
- Etc.

I'll pick “user experience using the device” (which is still a big topic)

Research question / criteria

- RQ (Stands for Research Question): Which of the two keyboards provides the better user experience?
- Here, we can define user experience to be:
 - Comfort whilst using the device (for long durations)
 - Workload whilst using the device

We might even just explicitly say

RQ: Are there any significant differences in user comfort whilst using either keyboard over a long period of time (e.g. 1 week of continuous usage)?

RQ: Are there any significant differences between the user workloads whilst using either keyboard?

Defining various variables / factors

Dependent variable:

- **Comfort** — questions measured on a 5-point Likert scale / qualitative feedback
- **Workload** — measured via a standardised questionnaire (e.g. the Nasa TLX questionnaire)

Note: We will cover more about capturing this data via questionnaires in the following weeks

Defining various variables / factors

Independent variable: the two keyboards

Task: A recreation of office work — simulate 50 minutes office work (so various tasks such as writing emails, replying to slack messages, searching the web, writing a report, etc.)

Conditions: Keyboard One, Keyboard Two

Defining various variables / factors

Experimental objects: the human participants

Control variables: the same task used for both conditions / all participants, the same room used for all participants, the same computer monitor / chair / etc. within the room

Defining various variables / factors

Random variables: variations in human characteristics (e.g. prior technological experience)

Confounding factors: participants prior experience typing (e.g. touch typists vs single finger typers)

Anything else?

- Still need to report how we intend to analyse our data (even at the experimental design stage)
- Still need some mention of the ethics of running the experiment (particularly if human participants are involved)
 - More on these topics next week

Any Questions?

Summary from this lecture

- The definitions of “Within/Between Subject Designs” and “Experimental Factors / Variables”
- Counterbalancing, randomisation
- Human characteristics and participants
- Confounding factors

- You need to know the definitions of all of these terms — these are the language used to describe / define experimental designs

Quick Break Before Part 2